INDEX

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S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1)	20/7/26	Experiment of study of various Network Commands.		28/8/26 (9)
2)	28/7/26	study of different types of network cables		4/8/26 (8)
3)	10/8/26	To study the packet tracer tool installation and user Interface		21/8/26 (8)
4)	17/8/26	To setup and configure a LAN using switch and ethernet cables		29/8/26 (9)
5)	18/8/26	Experiment of Packet Based tool: Wireshark	(9)	18/8
6)	7/9/26	Hamming code Concept	(9)	7/9
7)	8/9/26	Sliding window Protocol	(9)	8/9
8)	22/9/26	Live Hosts using Nmap scans	(9)	22/9

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
9)	25/9/26	Implementing subnetting in Cisco Packet	9	23/9
10)	28/9/26	Internetworking with routers in cisco packet	9	28/9
11)	01/10/26	RIP using Cisco Packet Tracer	9	1/10
12)	01/10/26	Implementation of RAW Sockets.	9	1/10
13)	29/9/26	End to End communication	10	29/9
14)	30/9/26	Ping Program	10	30/9

20/1/26

Exp 1:- Experiment of study of various Windows commands:- Network Commands

1) arp -a

ARP is short term of Address Resolution Protocol

O/P:- Interface 10.16.56.50 --- 0x9

Internet Address	Physical Address	Type
192.168.1.1	00-1A-2B-3C-4D-5E	Dynam
192.168.1.20	08-00-27-AB-CD-EF	Dynami

2) hostname

To display the Computer name

O/P: DESKTOP-PC01

3) ip config /all

To display complete network configurat

4) nbstat - a

To display NetBIOS information.

Sample o/p : Net BIOS Remote

Machine Name

Name	TYPE
DESKTOP-PC01	UNIQUE
WORKGROUP	GROUP

5) netstat

To display active network connec and listening ports

Sample output

Proto	local Address	Foreign Address
TCP	192.11E.1.10:5050	142.250.183.36
TCP	0.0.0.0:135	0.0.0.0

6) netstat -r

To display routing table

Sample o/p:-

Network Destination	Gateway
0.0.0.0	192.168.1.1

7) nslookup www.google.com

To find the IP address of a domain name

Sample o/p:-

Server: dns.google
Name: www.google.com
Address: 142.250.183.36

8) pathping google.com

To display the network path & packet loss

Sample o/p:-

Tracing route to google.com
Hop 1 192.168.1.1
Hop 2 10.10.10.1
Hop 3 google.com
computing statistics

9) Ping google.com

To test network connectivity with another host.

Sample o/p:-

Pinging google.com [142.250.183.36]
Reply from 142.250.183.36
Bytes=32
Time=15ms
TTL=118

5) route print

To display the IP routing table

Sample o/p :-

Network Destination	Network	Gateway
0.0.0.0	0.0.0.0	192.168.1.1
192.168.1.0	255.255.255.0	on-link

Linux Commands

1) IP Command :ip address show

To display the IP addresses of all network interface

Sample o/p :-

2: eth0
inet 192.168.1.10/24
inet6 fe80::a00:27ff:fe4e:66a1/64

2) ~~ip address add~~ ip address add 192.168.1.100/24 dev eth0

To assign an IP addresses to network interface

Sample o/p :-

IP address added successfully

3) ip address del 192.168.1.100/24 dev eth0

To remove an IP address from a network interface

sample o/p:-

IP address removed successfully

4) ip link set eth0 up

To enable a network interface

sample o/p: eth0: state up

5) ip link set eth0 down

To disable a network interface

sample o/p: eth0: down

6) ip route add default via 192.168.1.1 dev eth0

To add the default gateway

sample o/p:- Default route added Successfully

7) ip route add 192.168.2.0/24 via 192.168.1.1

To add a route to another network

sample o/p: Route added successfully

8) ip route delete 192.168.2.0/24 via 192.168.1.1

To delete a route from the routing table

sample o/p:- Route deleted Successfully

9) ip route get 10.10.1.4

To display the route to a destination

Sample o/p: 10.10.1.4 via 192.168.1.1 dev eth0

10) ifconfig

To display network interface

Sample o/p:

eth0: flags=4163<UP,BROADCAST,RUNNING

inet 192.168.1.10

11) mtr google.com

To display the network path & statistics

Sample o/p:-

HOST	Loss%	Avg
192.168.1.1	0.0%	2ms
google.com	0.0%	18ms

RESULT:-

Thus the various windows and Linux commands were studied and executed successfully

28/8/24

Topic: Study of different types of network cables

a) understand different types of network cable
Different type of cable used in networking
1) unshielded twisted pair cable
2) shielded twisted pair cable
3) coaxial cable
4) fibre optic cable

Cable type	category	maximum data transmission	Advantages/ Disadvantages	Applications
UTP	category 5	10Mbps	Advantages * cheaper in cost * Easy to install as they have a smaller Disadvantages * More prone to EMI	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet
	category 5e	1Gbps		Fast Ethernet Gigabit Ethernet
	category 6	10Gbps		
STP	category 6a	10Gbps	Advantages * shielded * faster than UTP * Less susceptible to noise and interference	Gigabit Ethernet, 10G Ethernet widely used in data centers

Cable Type	category	Maximum Data Transmission	Advantages / Disadvantages	Applications
STP	category	10GPS	Disadvantages * Expensive * Greater installation Effort	Gigabit Ethernet
coaxial cable	RG-6 RG-59 RG11	6-100MBPS	*High Bandwidth * Low bandwidth * versatile Disadvantages * Limited distance * cost *Size is Bulky	Speed of signal is Television network High speed internet connec
Fibre optics Cable	single mode multi mode	100GbPS	Advantages * High speed * High Bandwidth * High security * Long Distance Disadvantages * Expensive * Requires skilled installers	Maximum distance of fiber optics cable around

RESULT:- Thus the different types of network cables were explained successfully.

01/8/26

Exp 3- To study the packet tracer tool installation and user Interface overview

Introduction:-

* A simulator as the name suggests, simulates network device and its environment.

1) It allows you to model complex systems without the need for equipment
2) It is available for both the linux and windows desktop environments

Installing Packet Tracer:-

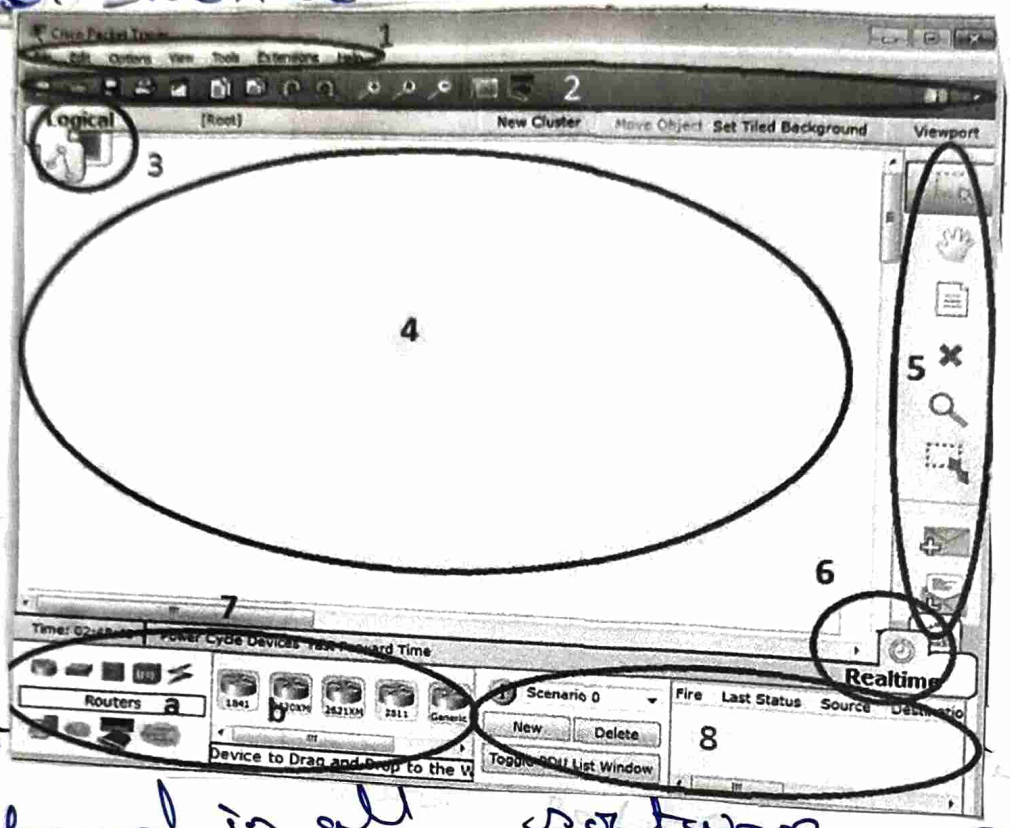
To download packet tracer go to https://www.netcad.com and login with your cisco Networking Academy then click on the packet Tracer graphical download the package

Installation in windows is pretty simple and straightforward the setup comes in a single life named packettracer-Setup 6.0.1.exe

Linux:-

Linux users with an ubuntu / Debian distribution should download the file for ubuntu and those using Redhat / centos must download the file for fedora

User Interface overview -



1) Menu bar - Menu bar in all software applications

2) Main toolbar - This bar provides shortcut icons to menu options that are commonly accessed such as open save etc....

3) Logical/physical workspace tabs - These tabs allow you to toggle between the logical and physical

4) Workspace - This is the area where topologies are created

5) common tools bar - This toolbar Provides controls for manipulating topologies such as select, move, layout place & Complex PDU

6) Real Time - These tabs are used to toggle between the real and simulation modes.

7) Network Component box - This component contains all of the network and end devices available with packet tracer

8) user - created box - users can create highly - customized packets to test their topology from this area and the results are displayed

9) Analyse the behaviour of network devices using cisco packet tracer simulator

1) From a network component

a) 4 Generic PCs and one HUB

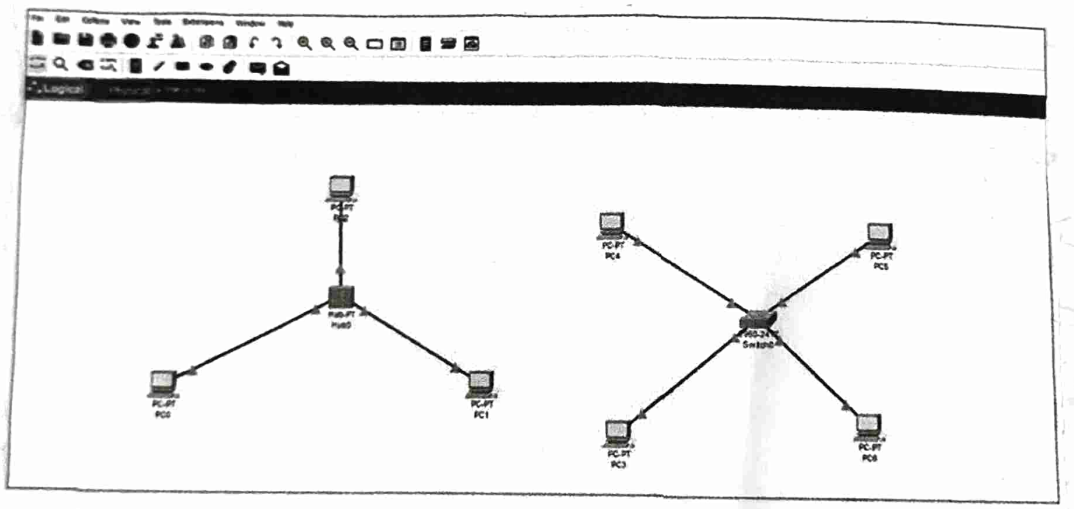
b) 4 Generic PCs and one switch

2) click on Connections:-

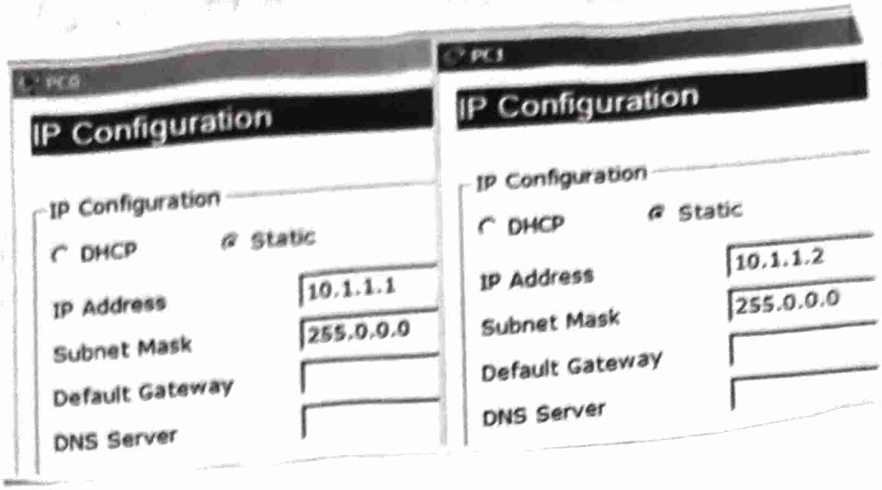
a) click on copper straight - through cable

b) select one of the PC and connect HUB using the cable

c) similarly connect 4 PCs to the switch using copper straight. through cable



3) click on the PCs (connected to hubs). go to the desktop tab. click on IP Config and enter an IP address and subnet mask.



click on the PDU from the common tool bar

a) Drag and drop it on one of PC and then drop it on another PC connected to HUB

4) observe the flow of PDU from source PC to destination PC by selecting the realtime mode of simulation

5) Repeat step 3 to step 5 for the PC connected

6) observe how HUB and switch are forwading the PDU and write in OBS & conclusion

RESULT:-

Thus the packet tracer tool installation has been downloaded and executed successfully

17/8/26

Expt 10 setup and configure a LAN using a switch and ethernet cables in your lab

AIM:- setup and configure a LAN using a switch and Ethernet Cable

LAN:-
A Local Area Network refers to a network that connects devices within a limited area such as an office building school or name. LAN connects devices to Promote collaboration and transfer between users such as computers, Printers, servers and switches.

How to set up a LAN:-

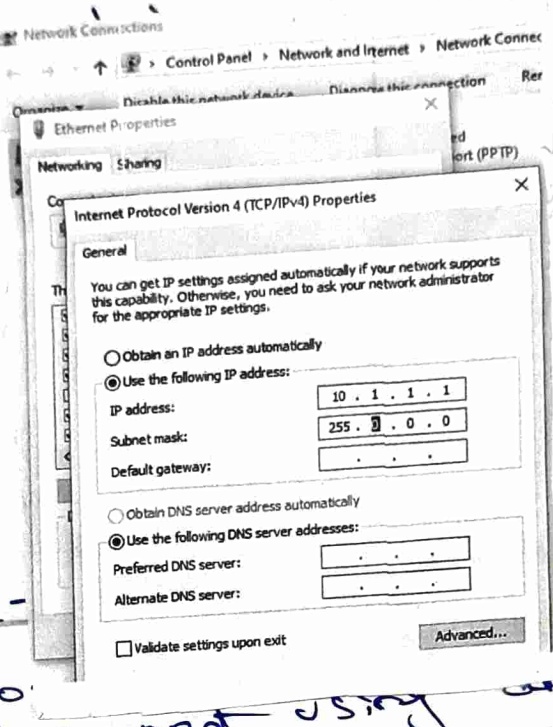
Step 1:- plan and design an appropriate network topology taking into account network requirement

Step 2:- You can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient your network.

Step 3:- Connect your computers to network switch via an Ethernet cable which is as simple as plugging in one end.

Step 4:-

* Log on to the client computer as administrator or as owner
* click network and Internet connection
* Right click Local Area Connection Eth -> properties -> select protocol



Step 5:-

* Co... switch and connect using ... ether to switch, an ethernet cable
* Log in to the web interface: open a web browser and enter IP address of the switch in address bar
* configure basic settings
* Assign IP address as 10.1.1.5

Step 6:- check the connectivity between switch and other machine by using ping command

Step 7:- select a folder -> go to properties -> click sharing tab

Step 8:- Try to access the shared folder from others computers of the network

RESULT:-

Thus the setup and config a LAN using switch and ethernet has been executed successfully.

29/9/24

Exp 5 - Experiment on Packet capture tool: Wireshark

- Packet sniffer
  - + Sniffs messages being sent/received from/by your computer
  - + stores and displays the contents of the various protocol fields in the messages

Diagnostic tools

- + Tcpdump
  - Eg: tcpdump -cnt host 10.129.41.2 -
- + Wireshark

Description:-

Wireshark:-

Wireshark a network analysis tool formed known as Ethereal, captures packets in real time and display them in human-readable format

- + capture network traffic
- + Decode Packet Protocols using dissectors
- + Watch smart statistics.

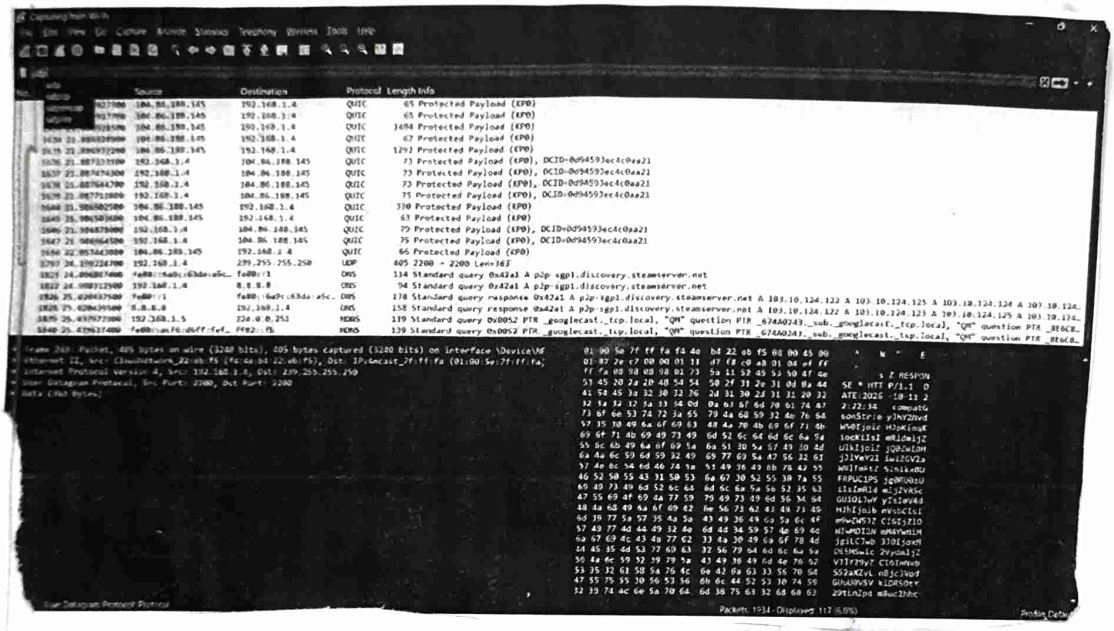
Getting wireshark:-

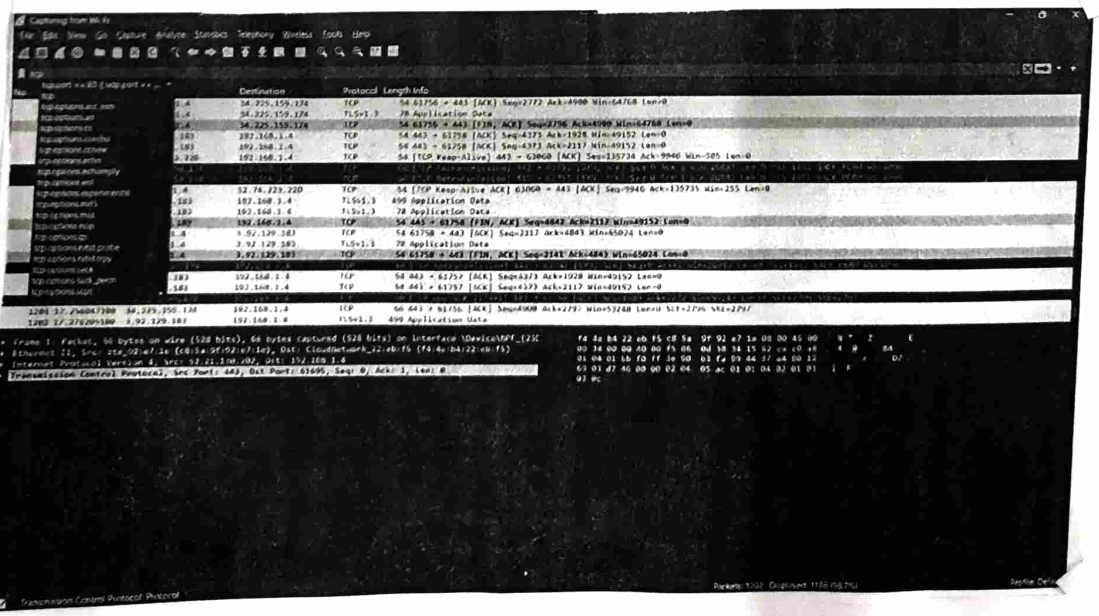
wireshark can be downloaded to windows or macos from its official website. For Linux or another unix system.

capturing Packets :-

After downloading and installing launch it and double-click the name of a network interface under capture

↳ capturing packets





The "Packet List" pane

The packet List pane displays all the packets in the current capture file. the "packet List" pane.

The "packet Details" pane

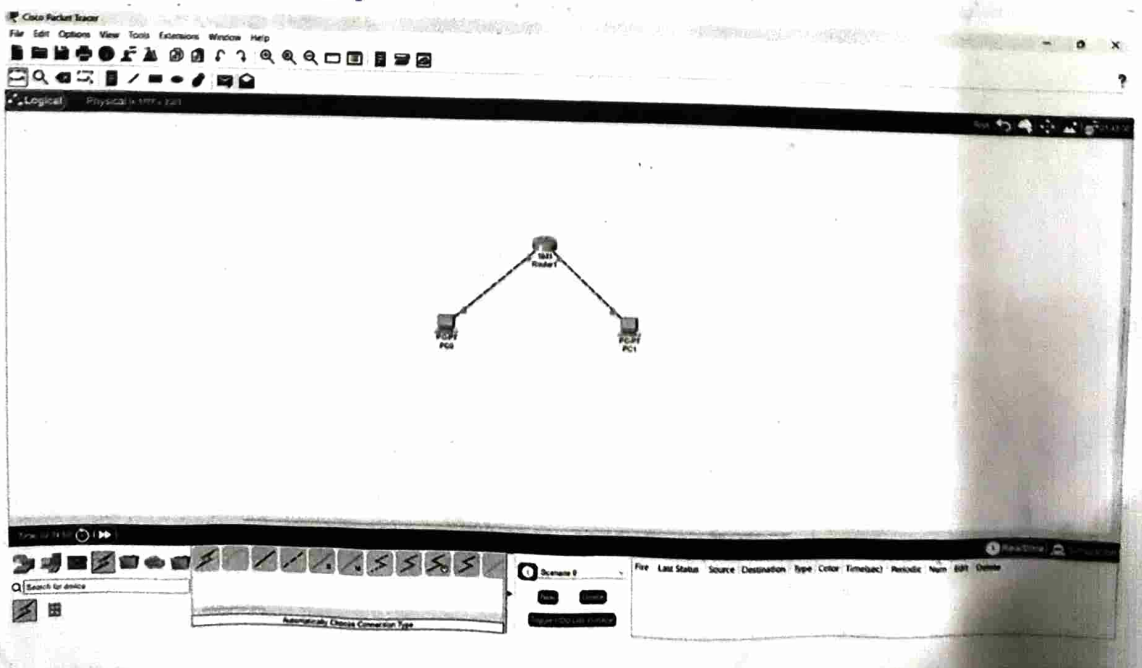
The packet details pane shows the current packet in a more detailed form.

The "packet Bytes" pane

The packet Bytes pane shows the data on the current packet

Color coding

Wireshark uses colors to help you identify the types of traffic at a glance. By default light purple is TCP traffic light blue is UDP traffic been

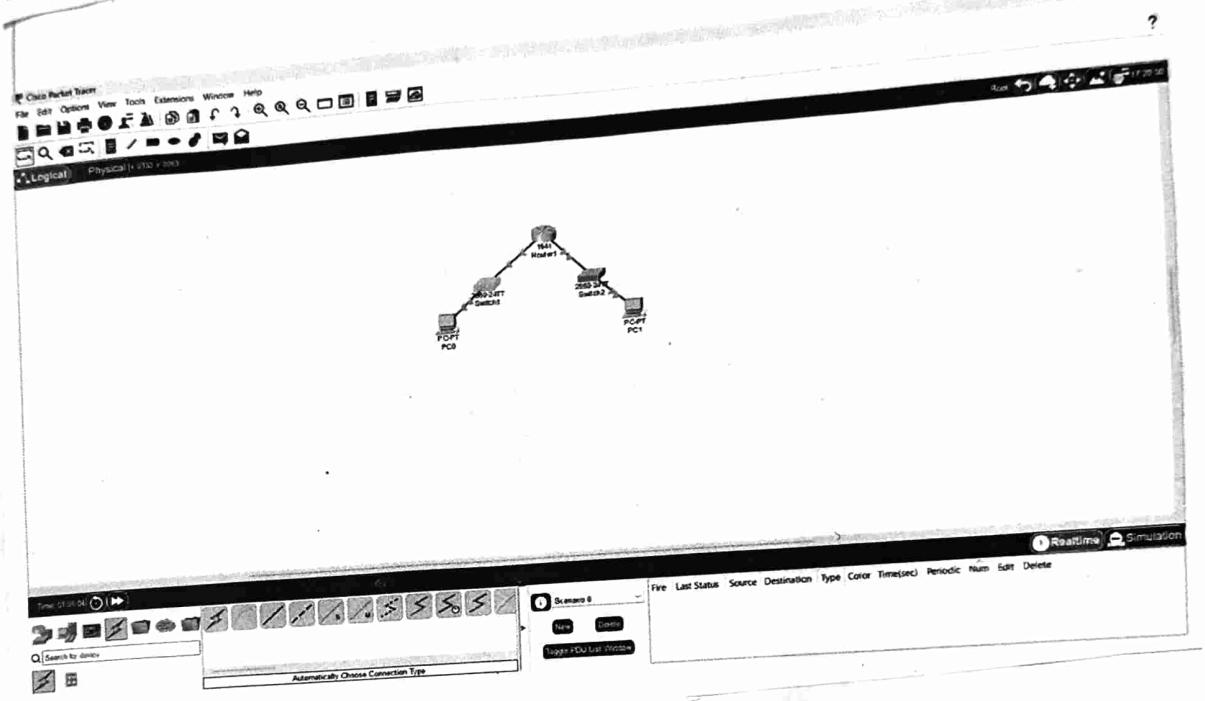


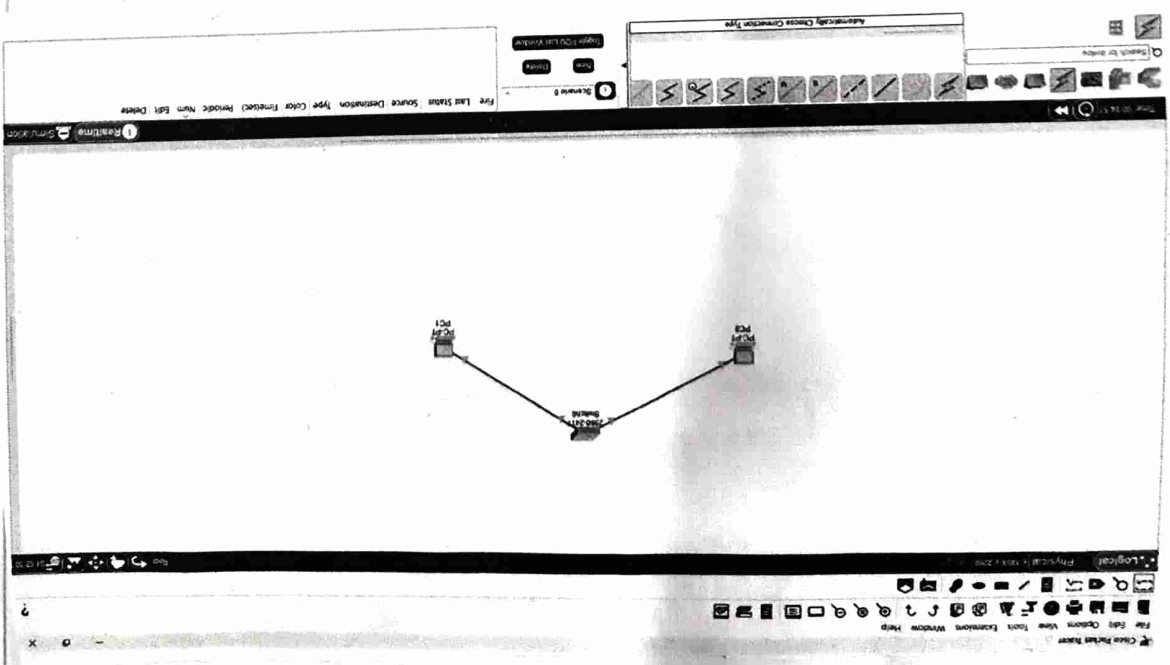


Filtering Packets

So you're are trying to inspect something specific such as the traffic a program sends when phoning home, it helps to close down all other application







Capturing and Analysing Packets using wireshark tool

Procedure

* Select Local Area Connection in Wireshark
* Go to capture option
* Save the packets

1) to display only TCP/UDP packets inspect the packets and

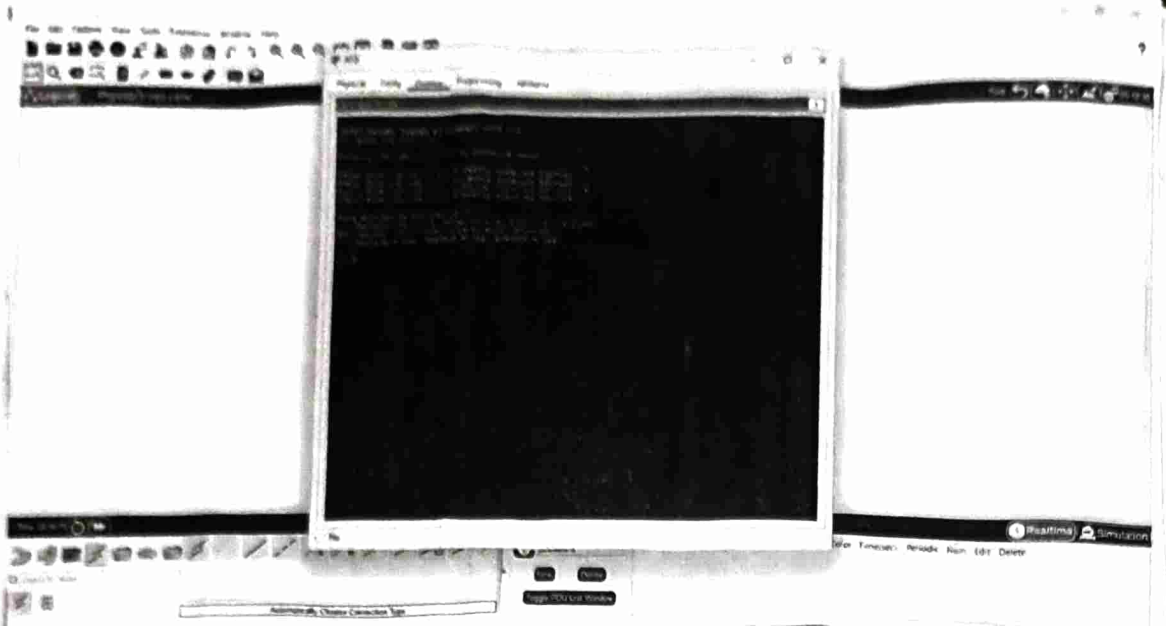
* Go to capture option
* select Local Area connection in wireshark
* Then click start capture
* save the packets.



2) create a filter to display only ARP pack

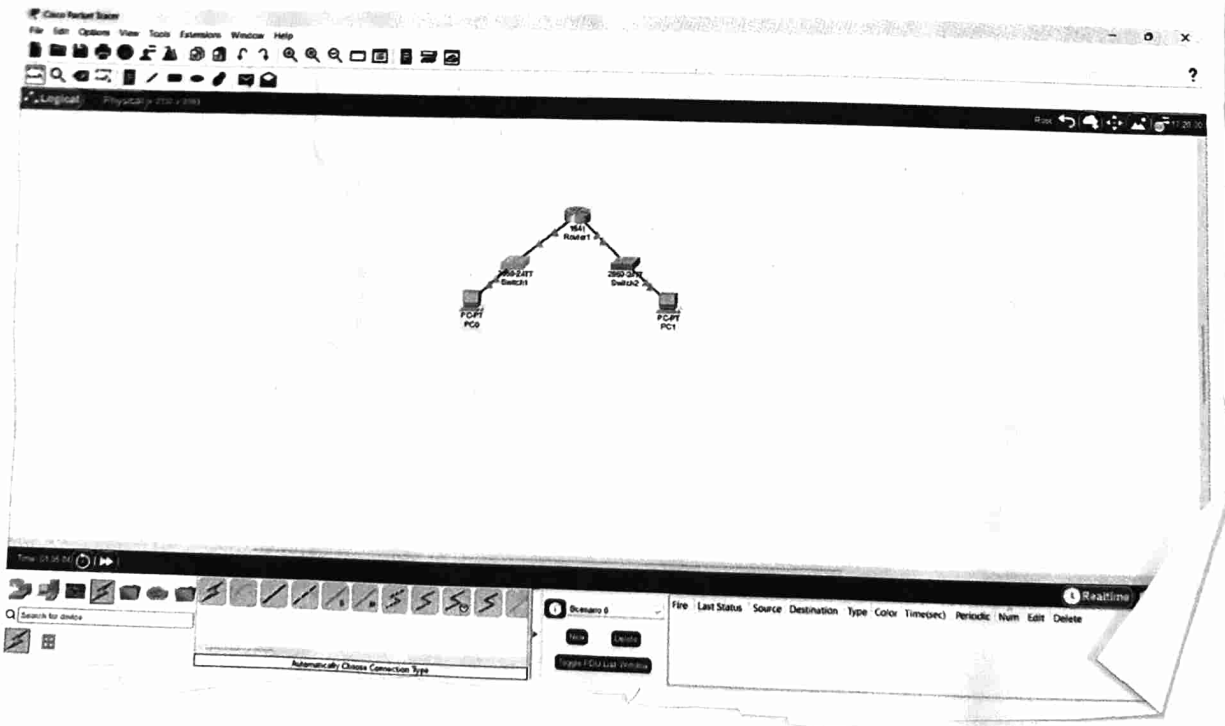
* Go to capture option
* select stop capture automatically after 100 packets
* Then click start capture
* save the packets

output:-



3) create a to display only DNS packets to provide the flow graph

+ Go to capture option
+ Search DNS packets in search box
+ To see flow graph click statistics



4) To display only HTTP pa
+ Go to capture option
+ then click start capture
+ search HTTP packets in search bar

RESULT:-

Thus the experiment on packet capture tool is used on wireshark has been executed successfully.

18/8/25

7/9/26 Exp6- HAMMING CODE CONCEPT

AIM:-

Need to implement error Detection using Hamming code concept.

Error correction at Data Link Layer:-

Hamming code is set of error-correcting codes that can be used to detect and correct the errors

CODING:-

```
data = input("Enter Data bits:")
m=len(data)
r=0
while 2**r<m+r+1:
    r+=1
n= m+r
hamming = ['0']*(n+1)
j=0
for i in range(1, n+1):
if (i & (i-1)) !=0
    hamming[i]= data[j]
    j+=1
for p in [1,2,4,8,16]:
    if p>n:
        break
```

```
count = 0
for i in range (1, n+1):
  if i != p:
    if hamming[i] == '1':
      count += 1
if count % 2 != 0:
  hamming[p] = '1'
else:
  hamming[p] = '0'
hamming_code = ''.join(hamming[1:])
syndrome = 0
for p in [1, 2, 4, 8, 16]:
  if p > n:
    break
  count = 0
  if received[syndrome] == '0':
    received[syndrome] = '1'
  else
    received[syndrome] = '0'
print("original Data:", data_received
```

I/P:- Enter data bits: 100011

O/P:- Hamming code: 00100001 1
No error detected

7/9/25 RESULT:- Thus the hamming code has been executed successfully

8/9/25 Exp 7- sliding window protocol

Aim:-

To implement flow control at data link layer using sliding window protocol

CODING:-

```
n=int(input("Enter no of frames:"))
w=int(input("Enter window size:"))

frames=[]
for i in range(1,n+1):
    frames.append(i)
print("frames:",frames)
i=0
while i<n:
    print("\n sending frames:")
    for j in range(i,min(i+w,n)):
        print("Frame",frames[j],'sent')
    ack=input("Acknowledgment received? (yes/no):")
    if ack=="yes":
        print("Acknowledgment received")

print("\n all frames sent successfully")
```

I/P:-

Enter no of frames: 6
Enter window size: 3

o/p:-

Sending frames

frame 1 sent
frame 2 sent
frame 3 sent

RESULT:-

Thus the sliding window has been executed successfully.

8/9/26

23/9/26 Expt:- To Discover Live Hosts using NMap

AIM:-

To discover LIVE Hosts using NMap SCANS (ARP, ICMP, TCP/UDP) on tryhackme platform ROOM

b) Configuration of WLAN using CISCO packet Tracer

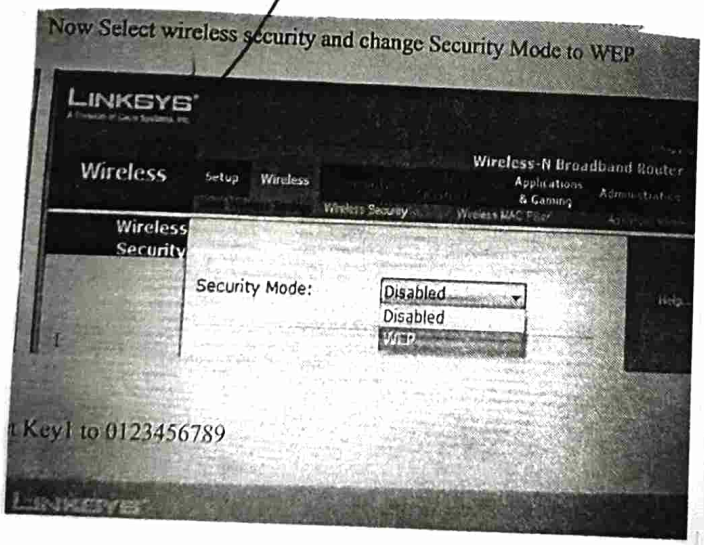
Perform following configuration

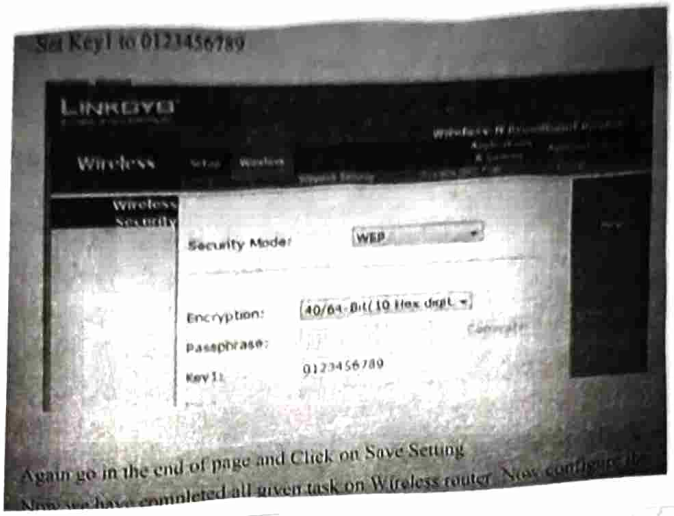
+ Configure static IP on PC and wireless Router

+ Set SSID to Mother network

Step :-

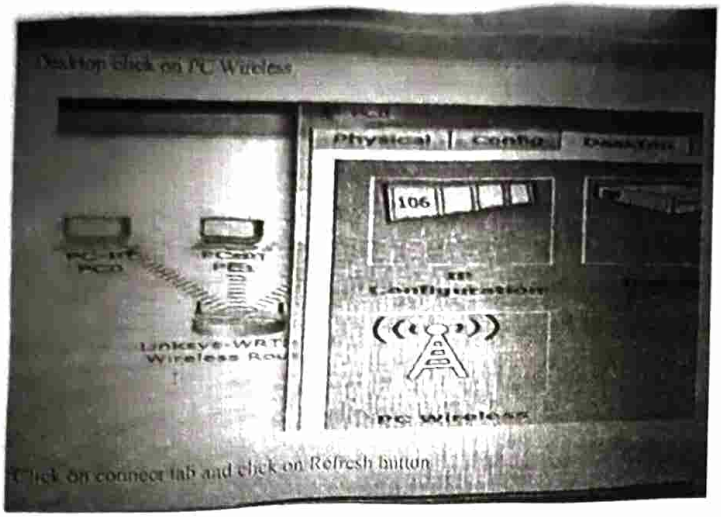
+ select Administration tab from top Menu, set username and password to admin and click on save setting



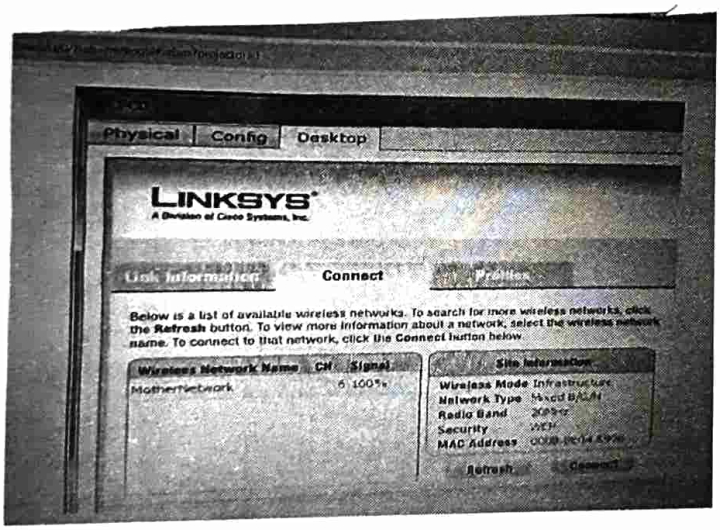


* Now we have completed all given task on wireless router

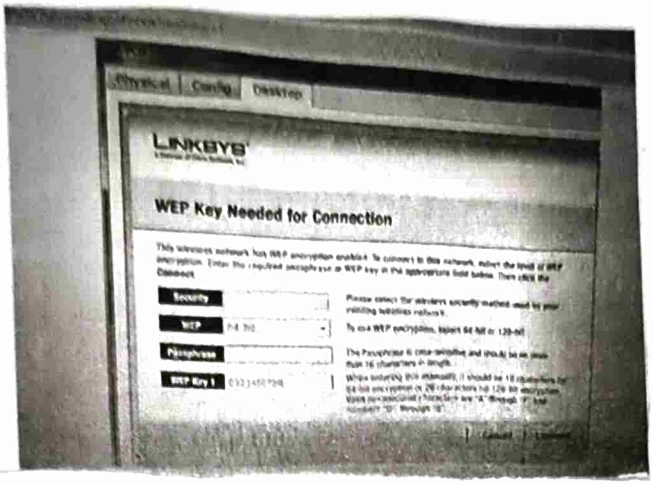
PC	IP	Subnet Mask	Gateway
PC0	192.168.0.2	255.255.255.0	192.168.0.1
PC1	192.168.0.3	255.255.255.0	192.168.0.1
PC2	192.168.0.4	255.255.255.0	192.168.0.1



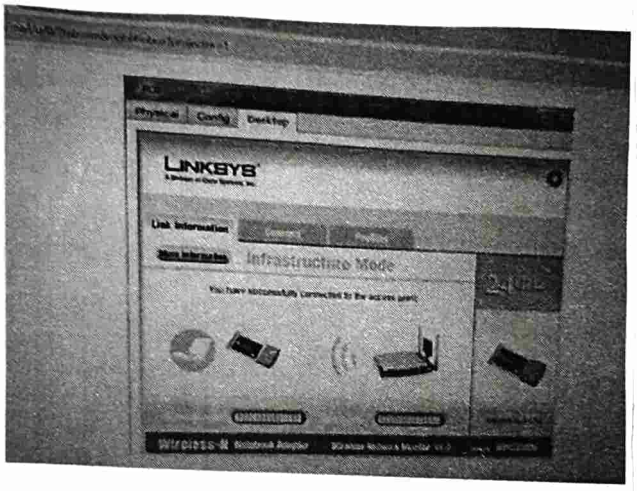
* Now its time to connect pc's from wireless router. To do click PC select Desktop click on PC wireless



* click on connect tab



* It will connect you with wireless router.



* Repeat same process on PC1 & PC2

RESULT:-

Thus discover of Live Hosts using Nmap scans has been completed successfully.

22/9

28/9/26

EXP 9 - Implementation of subnetting in cisco packet tracer

AIM:-

To implement subnetting in cisco packet tracer simulator

creating a network topology:-

The first step in implementing classless IP subnetting is to create a network topology in packet tracer.

Adding the devices:-

once we have created our network topology, we can add devices to it here we will be adding routers, switches and PCs.

Subnetting:-

To subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 addresses for end devices the switch and the router we can use a /27 subnet mask

The IP addressing for the network

* Router R1:-
* Gigabit Ethernet 0/0: 192.168.1.1
* Gigabit Ethernet 0/1: 192.168.2.1
* PC1: 192.168.1.11
* PC2: 192.168.1.12
* PC3: 192.168.1.13
* PC4: 192.168.1.14

the command-line interface for the router

enable
configure terminal
interface fastEthernet 0/0
ip address
no shutdown

Replace "{IP address}" and "{subnet mask}" with your desired IP address and subnet mask

The first interface, fast Ethernet 0/0, will be connected to the switch, while the second interface with IP addresses and subnet mask

These commands configure the switch to operate in access mode on its two ports.

Commands: enable
configure terminal
interface gigabit Ethernet 0/0
ip address {IP address} {subnet mask}

Testing the network:-

Now that our network topology is configured we can test the network. Open a command prompt on each PC and try to ping the other PC.

RESULT:-

Thus the implementation of subnetting in cisco packet tracer has been executed successfully.

28/9

Exp 10 - Interebranking
Packet Tracer

Aim:-
To design and configure a simple network using a router

Procedure:

Step 1
1) Select the router and open CLI
2) Press Enter to start configuring

Router command Line Interface:

Router> enable
Router# Config t
Enter configuration commands, one per line
CNTL/Z Router(config)# interface fast Ethernet 0/0
Router(config-if)# ip address 192.168.10.1

Step-2 (Configuring PCs)

1) Assign IP address
2) Select the PC, Go to the desktop and select IP configuration
3) Assign the default gateway of PC0 as 192.168.10.1
4) Assign the default gateway

Step-3

1) Connect Fast Ethernet 0 port of PC0 with fastEthernet 0/0 port of Router1 using a copper straight-through Cable

Router Configuration Table:-

Device name	IPadd	Subnet mask	IPadd	Subnet Mask
	(192.168.10.1)	255.255.255.0	192.168.20.1	255.255.255.0
Router1				

(b) Design and Configure an internetwork using wireless router, DHCP server and internet cloud

Step1:- Launch packet Tracer

Step 2: Build the topology

a) Add network devices to the workspace

To place a device on the workspace first choose a device type from the Device-Type selection box Then click on the desired device model from the device-specific selection box.

b) change display names of the network devices

To change the display names of the network devices click on the device icon on the packet tracer Logical workspace then click on the config tab in the device configuration window

c) Add the physical cabling between devices on the workspace

using the device selection box, add the physical cabling between devices on the workspace as shown in the topology diagram

Part 2:- Configure the network devices.

Step 1: Configure the wireless routers

a) create the wireless network router

click on the wireless router icon on the packet tracer Logical workspace to open the device configuration window

b) click on the save settings tab

a) colick the icon on the packet tracer Logical workspace and in the laptop Configuration windows selet the physical tab

Step 3:- Configure the PC

a) click on the PC icon on the packet tracer Logical workspace and selet the Desktop and selet the Desktop tab and then the IP configuration icon

Part 3: verify Connectivity

Step 1:- Refresh the IPv4 settings on the PC

a) click on the PC on the packet tracer logical workspace and then Select the Desktop tab of the PC configuration window

b) Test Connectivity to the cisco.com server from the PC

From the command prompt issue the command ping cisco.com
It may take a few seconds for the ping to return

RESULT :-

Thus the internetworking with routers in cisco packet tracer has been executed succes

11/10/24

Expt:- RIP using Cisco Packet tracer

AIM:-

To simulate RIP using Cisco Packet Tracer

* Assign IP address to pc's:-
  Double click pc's and click Desktop menu item and click IP Configuration
* Assign IP address to interfaces of routers:-
  Double click Routers and click CLI and press Enter key

```
Router>enable
Router# Configure
terminal
Enter Configuration Commands, one per line

Router(Config)# interface fastEthernet 0/0
Router(config-if)# ip-address 10.0.0.1
255.0.0.0 Router(config-if)
Router(config-if)# exit
Router(Config)#
```

no shutdown command will bring the interface up

exit command is used to return in global configuration mode

Router # show controllers serial 0/0/0
Interface Serial 0/0/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 2000000

Router 1:-

Router > enable
Router # configure
terminal
Enter configuration commands one per line
Router (config) # interface serial

Router 2:-

Router > enable
Router # configure
terminal
Enter configuration commands, one per line
Router (config) # interface fastEthernet 0/0
Router (config-if) # ip address 20.0.0.1

255.000 Router (config-if) #no shutdown

Router (config-if) #exit

Router (config-if) # interface Serial 0/0/0

Router (config-if) # ip address 192.168.1.2

255.255.255.252 Router #no shutdown

Configure RIP routing Protocol

Router0 (config) # router rip

Router0 (config-router) # network 10.0.0.0

Router0 (config-router) # network 192.168.

Router 0 :

* router rip command tell router to enable the RIP routing protocol
* network command allows us to specify the networks which we want to advertise.

Router1

Router (config)# router rip

Router1 (config-router)# network 192.168.1.0

Router 2

Router2 (config)# router rip

Router2 (config router) #network 20.0.0.0

RESULT:-

Thus internetworking with routers in cisco packet tracer has been executed successfully.

EXP 12 - RAW Sockets

AIM:-

To write a code using Raw sockets to implement packet sniffing

CODE:-

```
import socket
import struct

raw_socket = socket.socket(socket.AF_INE
socket.sock_D RAW
host = socket.gethostbyname(socket.get
                            name())
raw_socket.bind((host,0))

raw_socket.setsockopt(socket.IPPROTO_IP,
                      socket.IP_HDRINCL,1)

raw_socket.ioctl(socket.SIO_RCVALL,
                 socket.RCVALL_ON)

Print("Packet Sniffer started...")

Print("waiting for packets:\n")

while True:
    Packet, address = raw_socket.recv
                      from(65
    IP_header = packet[0:20]
```

```
iph = struct.unpack('!BBHHHBBH4s4s', ip_header)

version = iph[0] >> 4
Protocol = iph[6]
Source_ip = socket.inet_ntoa(iph[8])
destination_ip = socket.inet_ntoa(iph[9])
print("Source IP :", source_ip)
print("Destination IP:" destination_ip)
print("Protocol :" protocol)
print("-" * 40)
```

O/P :-

```
Packet sniffer started
Waiting for Packets
Source IP : 192.168.1.5
Des IP : 192.168.1.1
protocal : 6
```

RESULT :-

Thus the raw sockets have been implemented successfully.

10/25

29/9/26 EXP 13- End to End communication of transport layer

AIM:-

To implement an Echo server application using TCP Socket

CODE:-

ECHO SERVER

```
import socket
S = socket.socket()
s.bind(("localhost", 5000))
s.listen(1)
c, addr = s.accept()
msg = c.recv(1024).decode()
Print("client:" msg)
c.send(msg.encode())
c.close()
s.close()
```

ECHO CLIENT

```
import socket
s = socket.socket()
s.connect(("localhost", 5000))
msg = input("Enter message:")
s.send(msg.encode())
```

```
print('Echo:', s.recv(1024))
s.close()
```

I/P:-

Enter message: Hello server

O/P:-

Server

client connected

Received: Hello server.

RESULT:-

The Echo server using TCP sockets was studied and executed successfully.

30/9/26 Exp 14 Implement ... Ping Prog

AIM:-

to implement a ping program using python to check connectivity between source and destination host

CODE:-

```
import os
import time
host = input ("Enter host: ")
for i in range (4):
start = time.time()
result = os.system ("ping -n 1 " + host)
end = time.time()
if result == 0:
print ("Reply from host,
time = round ((end - start) * 1000, 2)
, "ms")
else
print ("Request Timed out")
time.sleep()
```

I/P :-

Enter host : 127.0.0.1

O/P :-

Reply from 127.0.0.1 Time=1.23ms

RESULT :-

The ping program has been executed successfully